INCH-POUND
MIL-DTL-24643/43G
27 November 2023
SUPERSEDING
MIL-DTL-24643/43F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPES LS2SJ, LS3SJ, AND LS4SJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, tin-coated, ASTM B286, for AWG sizes 14 and smaller. Coated or uncoated copper, ASTM B8, class B stranding, for AWG sizes 12 and larger.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for dimensions). Standard identification code applied by method 3.
Fourth	-	Two, three, or four conductors (see table I), as required, shall be cabled together with a lay in accordance with table I. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence for four conductor cables shall be consecutive.
Fifth	-	An optional binder.
Sixth	-	Braided shield of tin-coated copper.
Seventh	-	An optional binder.
Eighth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.035 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/43	Type and size	Conductors			Insulation wall thickness min. avg. (inch)	Lay of conductors max. (inches)	Overall diameter (inch)		Conductor resistance max. (ohms)
		No.	ASTM B286	ASTM B8			min.	max.	
-01UO	LS2SJ-22	2	22 – 19	--	0.016	1½	0.261	0.275	16.54
-02UO	LS2SJ-20	2	20 – 19	--	0.016	1½	0.273	0.290	10.16
-03UO	LS2SJ-18	2	18 – 19	--	0.016	1½	0.295	0.310	6.47
-04UO	LS2SJ-16	2	16 – 19	--	0.016	2½	0.309	0.325	5.02
-05UO	LS2SJ-14	2	14 – 19	--	0.016	2½	0.337	0.350	3.17
-06UO	LS2SJ-12	2	--	12 (class B)	0.024	3	0.417	0.430	1.72
-07UO	LS2SJ-11	2	--	10 (class B)	0.025	3	0.447	0.460	1.085
-08UO	LS2SJ-9	2	--	9 (class B)	0.031	4	0.525	0.545	0.86
-09UO	LS2SJ-7	2	--	7 (class B)	0.032	4	0.600	0.615	0.54
-10UO	LS3SJ-22	3	22 – 19	--	0.016	1½	0.271	0.285	16.54
-11UO	LS3SJ-20	3	20 – 19	--	0.016	1½	0.284	0.300	10.16
-12UO	LS3SJ-18	3	18 – 19	--	0.016	1½	0.308	0.325	6.47
-13UO	LS3SJ-16	3	16 – 19	--	0.016	2½	0.323	0.340	5.02
-14UO	LS3SJ-14	3	14 – 19		0.016	2½	0.353	0.370	3.17
-15UO	LS3SJ-12	3	--	12 (class B)	0.024	3	0.440	0.455	1.72
-16UO	LS3SJ-9	3	--	9 (class B)	0.031	4	0.594	0.620	0.86
-17UO	LS4SJ-20	4	20 – 19	--	0.016	1½	0.303	0.320	10.16
-18UO	LS4SJ-16	4	16 – 19	--	0.016	2½	0.346	0.360	5.02
-19UO	LS4SJ-14	4	14 – 19	--	0.016	3	0.380	0.395	3.17

NOTE:

1. Size designation does not correlate to conductor size for all types and sizes.

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1,200
Conductor to overall shield	600
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50

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Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2SJ, LS3SJ, and LS4SJ (all sizes)	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-059)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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